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Mapping wide-area land subsidence from groundwater use in the North China plain by machine learning-based InSAR adjustment

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ABSTRACT

Since 2014, a series of China's water resource management policies have been implemented to mitigate groundwater-induced land subsidence in the North China Plain (NCP). While previous studies have demonstrated the benefits of Synthetic Aperture Radar interferometry (InSAR) in providing policy-relevant insights into the spatio-temporal dynamics of subsidence and groundwater recovery, most have focused on localized regions, leaving the long-term impact of these measures across the entire NCP insufficiently evaluated. A key challenge is the variation of long-wavelength errors in each SAR frame, which results in inconsistencies in the subsidence velocity field over large areas. To address this issue, this paper proposes a machine learning-based adjustment approach for routinely wide-area subsidence mapping and then fully evaluating land subsidence and associated groundwater depletion in the NCP from the end of 2014 to 2022. The novelty of this method lies in the adaptive selection of the optimal model for each SAR frame, which minimizes the varying long-wavelength errors, rather than relying on a unified model for all SAR frames as commonly used in state-of-the-art approaches. Additionally, we mitigated the difference of InSAR measurement in the overlap regions between consecutive tracks caused by the varying incidence angles by incorporating GNSS data and a plate motion model. Using synthetic and real Sentinel-1 data, we validated the performance of the proposed method against prevalent approaches through an independent GNSS validation dataset, demonstrating accuracy improvements from 3.8-17.5 mm/yr to 2.0 mm/yr. The results indicated that approximately 56,882 km^2 of the NCP area experienced land subsidence greater than 20 mm/yr. The central alluvial and coastal plains were the primary areas of subsidence, with a maximum cumulative subsidence of up to 2 m. The average subsidence velocity peaked in 2018 at 38.5 mm/yr. Subsidence has been alleviated after 2021. Our results revealed the lower bound of groundwater loss from the confined aquifer in the NCP, totaling 24.9 billion m^3 between the end of 2014 and 2022. Of this total, 20.2 billion m^3 (81 %) was lost from October 2014 to the end of 2020, with the loss decreasing to 4.7 billion m^3 (19 %) during the period from January 2021 to December 2022. This study provides new evidence for China's groundwater management practices in addressing land subsidence in the NCP.

1. Introduction

Wide-area land subsidence induced by groundwater pumping has been observed globally and primarily results from the compaction of sediments as pore pressure in an aquifer system decreases (Hasan et al., 2023). This sustained subsidence can lead to reduced groundwater storage capacity, saltwater intrusion, aquifer salinization, damage to surface infrastructure, and the formation of earth fissures (Kinzelbach et al., 2022; Yang et al., 2018). Despite its significance, groundwater-related subsidence continues to occur in many regions of the world (Hasan et al., 2023).

One region under significant stress is the North China Plain (NCP). At broader scales and coarser resolutions, Gravity Recovery and Climate Experiment (GRACE) observations have underscored that the NCP has experienced long-term groundwater depletion over the past decades (Xu et al., 2021), characterized by shallow aquifer depletion in piedmont plains and deep groundwater depletion in the central plains (Huang et al., 2015). Subsidence caused by groundwater overexploitation was

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first observed as early as the 1920s (Guo et al., 2015). Nearly half a century later, the NCP has experienced unprecedented economic growth and urban expansion. Major cities such as Beijing, Tianjin, and those in Hebei and Shandong provinces rely heavily on groundwater to meet agricultural and industrial demands, leading to groundwater declines and the continuous expansion of land subsidence across approximately 120,000 km^2 (Ye et al., 2016). By the end of 2013, nearly 42.3 % of the area, totaling 48,500 km^2, was subsiding at rates greater than 20 mm/yr (Gong et al., 2018; Ye et al., 2016).

Spaceborne synthetic aperture radar interferometry (InSAR) provides an effective means to monitor groundwater-induced land subsidence with wide coverage and millimeter accuracy, and has been extensively applied in regions such as the USA, China, Iran, Indonesia, and Bangladesh (Ao et al., 2024; Haghshenas Haghighi and Motagh, 2024; Hakim et al., 2023). A few studies have used InSAR to assess groundwater storage depletion in confined aquifer systems (Bai et al., 2022; Haghshenas Haghighi and Motagh, 2024; Khorrami et al., 2023; Peng et al., 2022; Qu et al., 2023; Smith and Majumdar, 2020). However, practical assessments at regional or national scales remain challenging, as atmospheric delays, orbital errors, and the relative nature of LOS measurements often cause discontinuities between InSAR velocity frames, thereby limiting the accurate evaluation of wide-area land subsidence and groundwater loss.

Several approaches have been presented to address this issue, falling into three categories. The first approach is to reference the velocity fields of all SAR frames to a known Terrestrial Reference Frame (TRF) using GNSS (Global Navigation Satellite Systems) continuously operating reference station (CORS). For example, the European Ground Motion Service utilized 3800 stations on a regular 50 km grid to correct InSAR long-wavelength errors on a European scale (Dehls et al., 2022). The second approach is to minimize velocity field discrepancies of tie points in the overlap region between consecutive frames through least-square adjustments (Ou et al., 2022). While a polynomial model can be empirically chosen to fit long-wavelength bias, the polynomial order must be small enough to prevent overfitting and underestimating low-frequency deformation signals, yet large enough to accurately model more complex long-wavelength bias at some frames. The third approach aims to reduce the impact of incidence angle variations within the overlap regions between adjacent SAR tracks. As demonstrated by Fuhrmann and Garthwaite (2019), for Sentinel-1 TOPS data, a three-dimensional motion of 10 mm/yr in both the horizontal and vertical directions can result in an apparent LOS velocity difference of approximately 3 mm/yr at tie points within the overlap areas. The assumption of negligible horizontal displacement is commonly applied in most studies (Haghighi and Motagh, 2019; Wang, Feng, et al., 2024). Nevertheless, when sufficient SAR data are available, jointly using ascending and descending observations to derive the vertical deformation component provides a more robust and physically meaningful solution.

For the NCP spanning approximately 140,000 km^2, the distribution of GNSS data is sparse, and only ascending SAR data provides full coverage of the area. This makes applying state-of-the-art methods for wide-area land subsidence mapping impractical. In recent years, the majority of studies have focused on certain regions within the NCP. One such study analyzed the tri-decadal evolution of land subsidence in the Beijing Plain by integrating multi-source SAR data, identifying major subsidence funnels with a maximum vertical displacement of 1.98 m between 1992 and 2022 (Dong et al., 2023). One study analyzed the coupled processes of ground subsidence and groundwater level rebound in Tianjin using leveling, GNSS, and Sentinel-1 data collected between 2015 and 2022 (Su et al., 2023). Other studies investigated the impact of subsidence on infrastructure, such as airports, subway lines, and high-speed railways in Beijing and Tianjin, using high-resolution SAR data from 2010 to 2019 (Chen et al., 2021; Dai et al., 2020). Additionally, two studies combined land subsidence data from ALOS PALSAR and Envisat ASAR and in-situ field data to quantify total groundwater storage depletion from 1960 to 2010. These studies revealed that approximately 87 % of the total groundwater storage depletion in Cangzhou, located in the southeastern part of Hebei Province, is irreversible, resulting in permanent loss of groundwater storage capacity (Bai et al., 2022; Jiang et al., 2018). Another study utilized Sentinel-1 data stacks to investigate the subsidence along the Middle Route canal of South-North Water Diversion (SNWD) project since the end of 2014 (Dong et al., 2021), which has transferred a total of **29.7 billion m^3** of water to northern China over a five-year period, significantly reducing the NCP's dependence on groundwater pumping.

With the implementation of the SNWD project and the release of measures such as seasonal land fallowing, water resources fee and tax, and groundwater extraction bans (Kinzelbach et al., 2022; Long et al., 2025; Xu et al., 2025), the long-term impact of these policies on the global scale of land subsidence and associated groundwater depletion in the NCP has not yet been fully evaluated. This study aims to map long-term widespread land subsidence across the entire NCP at a fine spatial resolution and provide detailed insights into the associated groundwater depletion from the end of 2014 to 2022. The technical focus is on addressing challenges of spatial discontinuity in wide-area interferometry. Unlike state-of-the-art methods that rely on a unified model, we propose a machine learning-based approach to accommodate diverse long-wavelength bias in each SAR frame, resulting in consistent wide-area velocity field. Also, we account for the horizontal motion of tectonic plates across the Earth's surface in wide-area scenes to reduce the dispersion of subsidence at the overlap regions between consecutive tracks.

Using the newly developed methodology, we first utilize 2251 Sentinel-1 images and 20 CORS stations to map consistent land subsidence at a resolution of approximately 100 m. Next, we validate the performance of the proposed method against state-of-the-art approaches using both synthetic and real data. We then analyze the land subsidence patterns across the NCP. Additionally, we discuss the computational efficiency and uncertainty of the proposed method. Also, we discuss the annual temporal evolution of subsidence by incorporating hydrological data. Furthermore, we estimate groundwater storage loss from the confined aquifer based on InSAR observations and evaluate the variation in loss before and after 2021. Finally, we compare our wide-area subsidence results with those obtained from localized investigations in previous studies.

2. Materials

2.1. Study area

The NCP is located to the east of the Taihang Mountains, south of the Yanshan Mountains, north of the Yellow River, and west of the Bohai Gulf (Fig. 1a). It is the largest alluvial plain in China, covering an area of approximately 140,000 km^2. The plain includes Beijing and Tianjin municipalities, along with the northern plains of the Yellow River in Shandong and Henan Provinces. This region experiences a semi-humid to semi-arid continental monsoon climate with an annual precipitation of 500-600 mm and evaporation of 1100–1800 mm (Xue et al., 2019). According to the gauge-based gridded precipitation dataset for the Chinese mainland, the annual precipitation increased to 640 mm during the period from 2014 to 2022 (Han et al., 2023).

As one of the largest aquifer systems in the world, the NCP features three distinct hydrogeological settings: the piedmont plain with major alluvial fans, the central alluvial plain, and the coastal plain near the Bohai Sea (Fig. 1c). The thickness of the unconsolidated Quaternary sediments across these settings ranges from 200 to 600 m. From the mountainous escarpments that bound the aquifer system to the coastal areas, the aquifer systems transition from a single aquifer predominantly made up of sandy gravel and pebble to multiple layers separated by gravel, coarse to fine sands, silt and clay (Guo et al., 2015). Below most of the central alluvial and coastal plains, the sequence of aquifers

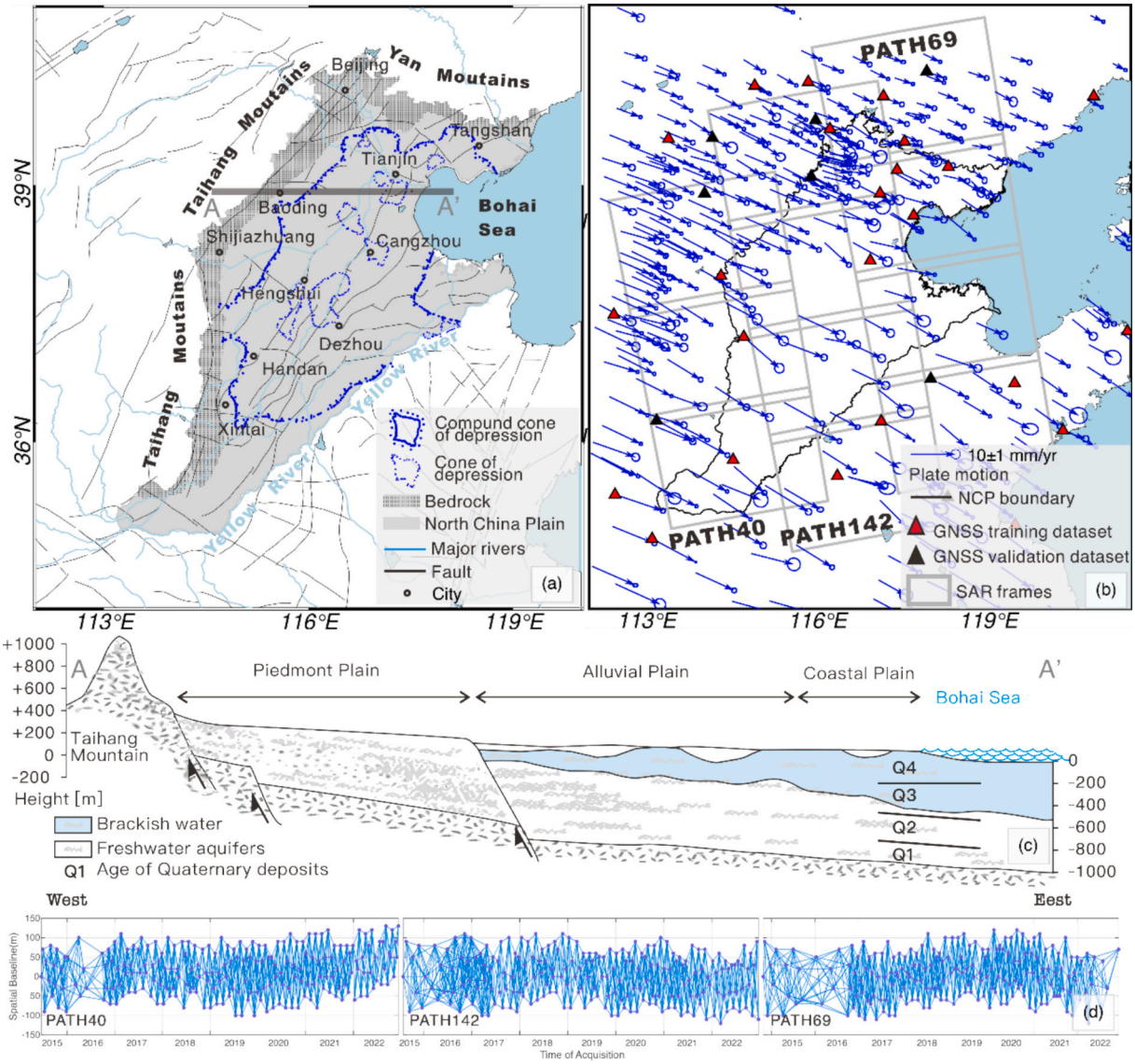


Fig. 1. (a) Sketch map of the NCP. (b) Geodetic dataset. The triangles represent the GNSS stations and the footprint in gray represents the Sentinel-1 image coverage. The blue arrow shows the horizontal plate motion model under Eurasian plate. (c) Hydrogeological cross section from Baoding area to Bohai bay of the NCP (modified from Foster et al., 2004). (d) Temporal-spatial baseline configurations of SAR time-series. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

includes the Holocene aquifer (depth of 10 to 50 m), the Late Pleistocene aquifer (depth of 110-170 m), the Middle Pleistocene aquifer (depth of 250 to 310 m) and the Early Pleistocene aquifer (depth of 350 to 550 m) (Guo et al., 2015; Xue et al., 2019). In the piedmont plain and associated alluvial fans, both the Holocene aquifer and Late Pleistocene aquifer contain fresh groundwater, which is replenished by recharging from the upper side of the plain. These two aquifers have already been jointly exploited, with the Late Pleistocene aquifer being confined. In contrast, the central alluvial and coastal plains are characterized by an extensive overlying brackish aquifer. Groundwater withdrawals in these areas primarily target the Middle and Early Pleistocene aquifers, raising concerns about the significant freshwater replenishment to the deep aquifer (Foster et al., 2004).

It has been reported that the overexploitation of deep groundwater for urban water supply and agricultural irrigation is the main driver of rapid land subsidence in the central alluvial and coastal plains (Kinzelbach et al., 2022). The long-term pumping has led to a significant decline in the pressure head, with the potentiometric surface dropping by as much as 75 m by 2010 (Guo et al., 2015). This continuous groundwater level decline has resulted in the formation of seven local depression cones, distributed across several cities including Langfang, Tianjin, Cangzhou, Hengshui, and Dezhou, which have expanded and merged to form a large compound cone of depression, covering an area of 72,787 km^2 (Fig. 1a).

2.2. Data

We collected three ascending Sentinel-1 tracks with 11 SAR frames, totaling 2251 time-series acquisitions from October 2014 to December 2022. The CORS network, consisting of 20 stations sourced from the National Earthquake Data Center (CMONOC) (https://www.eqdsc.com/), was utilized as control points for the InSAR block adjustment. Additionally, 305 horizontal secular velocities in the North China/Capital Circle, obtained from Wang and Shen (2020), were employed to estimate the Euler rotation vector and subsequently the horizonal displacement of the tectonic plate. The distributions of SAR and GNSS stations are illustrated in Fig. 1b. The details of major data sources are listed in Table 1.

Table 1
Major data sources used in the study.

Data	Details					
SAR	Orbit	Path	Mode	Resolution	Time	Image number
Sentinel-1	ascending	40	IW	5×20m	2015.06–2022.12	615
	ascending	142	IW	5×20m	2015.07–2022.12	800
	ascending	69	IW	5×20m	2014.10–2022.12	836
GNSS	Number	Time	Obs. frequency		Source	
CMONOC	20	2014–2022	daily		National Earthquake Data Center	
SSV*	305	1995–2016	< 7 days		(Wang and Shen, 2020)	
Hydrology data	Number	Time	Source			
Unconfined well	1284	2015–2021	Institute of China Geological Environment Monitoring			
Confined well	569	2015–2021				
Brackish water	/	/	GeoCloud Service in China Geological Survey			
Precipitation	/	2014–2022	(Han et al., 2023)			
Groundwater yield	/	2014–2022	National Hydrological Statistical Bulletin			
SNWD-diverted water to NCP	/	2014–2022				

* SSV denotes site secular velocities.

We also collected the hydrological dataset for the NCP, which includes in-situ groundwater levels from 2014 to 2021 as reported in the Groundwater Level Yearbook compiled by the Institute of China Geological Environment Monitoring (Institute of China Geological Enviornment Monitoring, 2021). Additionally, we utilized a new gridded precipitation dataset for mainland China (Fig. S1 in the supplementary material), featuring daily temporal resolution and a spatial resolution of 0.1° × 0.1° (Han et al., 2023). We also accessed data on the distribution of brackish water from the GeoCloud service provided by the China Geological Survey. Furthermore, the annual groundwater yield and the water volume transferred by the SNWD project were obtained from the National Hydrological Statistical Bulletin issued by the Ministry of Water Resources of the People's Republic of China (http://www.mwr.gov.cn/sj/tjgb/swxytjnb/).

2.3. Pre-processing

In InSAR data analysis, we first used the Sentinel Precise Orbit Determination Service and TanDEM-X 90 m Digital Elevation Model to execute geometrical co-registration (https://tandemx-science.dlr.de/). Then the enhanced spectral diversity technique was employed for azimuth co-registration. To mitigate the possible decorrelation between consecutive bursts, a temporal network with the maximum temporal baseline 60 days in Fig. 1d was used to invert the time-series azimuth shifts for each track (Jiang, 2020). After achieving azimuth co-registration accuracy better than 1/1000 pixel, all time-series single look complex images were resampled to align with the reference geometry. During interferometric processing, 20 × 4 looks were used to generate interferogram series with an approximate resolution of 100 m. Goldstein filtering was applied to mitigate stochastic noise in each interferogram, followed by minimum-cost-flow phase unwrapping. To mitigate tropospheric delays, we utilized products from the Generic Atmospheric Correction Online Service (GACOS) (https://www.gacos.net/). Finally, the velocity field was derived frame by frame using the LiCSBAS software (Morishita et al., 2020).

3. Brief review to adjustment for wide-area interferometry

3.1. Projection of satellite line-of-sight measurements

What InSAR measures is the one-dimensional projection of the three-dimensional ground target's movement along the LOS direction. The relationship between the LOS velocity v_{LOS} and the three-dimensional components (i.e., Up-Down v_{UD}, East-West v_{EW}, North-South v_{NS}) can be expressed as,

$$v_{LOS} = v_{NS}\cdot\sin\theta\cdot\sin\alpha - v_{EW}\cdot\sin\theta\cdot\cos\alpha + v_{UD}\cdot\cos\theta \quad (1)$$

where θ is the satellite incidence angle varying from 29° in near range to 46° in far range for TOPS mode. α is the heading angle fixed at −15° for ascending and − 166° for descending data, respectively. Typically, when both ascending and descending LOS velocities (i.e., v_{LOS}^{asc} and v_{LOS}^{des}) are available, one can invert Up-Down and East-West velocities by ignoring the North-South velocity:

$$\begin{cases} v_{LOS}^{asc} = -\sin\theta^{asc}\cdot\cos\alpha^{asc}\cdot v_{EW} + \cos\theta^{asc}\cdot v_{UD} \\ v_{LOS}^{des} = -\sin\theta^{des}\cdot\cos\alpha^{des}\cdot v_{EW} + \cos\theta^{des}\cdot v_{UD} \\ v_{NS} = 0 \end{cases} \quad (2)$$

where θ^{asc} and α^{asc} represent the incidence angle and heading of the ascending track, while θ^{des} and α^{des} represent those of the descending track. When ascending and descending data are not simultaneously available, an alternative is to project velocity from LOS to vertical direction using $v_{UD} = \frac{v_{LOS}}{\cos\theta}$, assuming v_{EW} and v_{NS} equal zero. This is valid when vertical displacement predominates in the observed scenes.

3.2. InSAR Block adjustment

The block adjustment process uses tie points (marked as circles in Fig. 2 Step 2) and/or control points (marked as triangles in Fig. 2 Step 2) to constrain the stitching of image frames by minimizing residual errors. This technique enhances the accuracy of the image alignment, ensuring the consistent velocity field (Wang, Gong, et al., 2024). Assuming that the discontinuities in velocities between adjacent SAR frames are primarily caused by long-wavelength bias within each frame, and that such bias can be modeled using a two-dimensional linear polynomial, then for an arbitrary tie point p with coordinates $\left(x_p, y_p\right)$ located in the overlap region, we can establish the constraint by $v_p^i - \left(\beta_0^i + \beta_1^i x_p + \beta_2^i y_p\right) = v_p^j - \left(\beta_0^j + \beta_1^j x_p + \beta_2^j y_p\right) + \varepsilon_p$, where $\left[\beta_0^i, \beta_1^i, \beta_2^i\right]$ and $\left[\beta_0^j, \beta_1^j, \beta_2^j\right]$ are the model parameters of the ith and j^{th} frame respectively. v_p^i and v_p^j show the observed velocities of adjacent SAR frames at position p in the overlap region. ε_p is the residual. When GNSS data is available, one can use the GNSS velocity located at q in the ith frame $v_q^{i,GNSS}$ as a control point, and

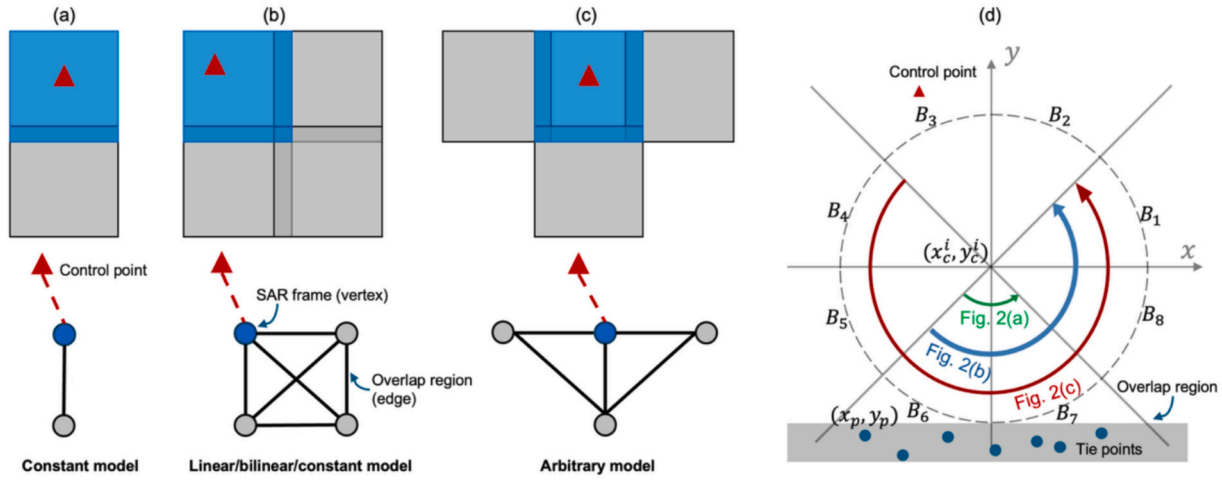


Fig. 2. Adjacent frames (gray) relative to the central frame (blue) and their corresponding graph structures ((a)-(c)). The red triangle indicates the GNSS control point. (d) The strategy used to define feasible sub-models for *K*-fold cross-validation. The local coordinate system centered on the blue frame quantifies the tie point distribution using positive angles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

we have $v_q^i - \left(\beta_0^i + \beta_1^i x_q + \beta_2^i y_q\right) = v_q^{i,GNSS} + \varepsilon_q$. Hence, for all p tie points in the overlap region and q GNSS control points at different frames, the following linear system holds,

$$\begin{bmatrix} v_n \\ v_e \end{bmatrix}_p = r_e \begin{bmatrix} \sin\lambda & -\cos\lambda & 0 \\ -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \end{bmatrix}_p \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \tag{4}$$

$$\underbrace{\begin{bmatrix} \begin{bmatrix} v_1^i - v_1^j \\ v_2^i - v_2^j \\ \\ v_p^i - v_p^j \end{bmatrix}_{p\times 1} \\ \begin{bmatrix} v_{p+1}^i - v_{p+1}^{i,GNSS} \\ \\ v_{p+q}^j - v_{p+q}^{j,GNSS} \end{bmatrix}_{q\times 1} \end{bmatrix}}_{v_{(p+q)\times 1}} + \underbrace{\begin{bmatrix} \begin{bmatrix} -1 & -x_1 & -y_1 & 1 & x_1 & y_1 \\ -1 & -x_2 & -y_2 & 1 & x_2 & y_2 \\ & & & & & \\ -1 & -x_p & -y_p & 1 & x_p & y_p \end{bmatrix}_{p\times 6} \\ \begin{bmatrix} -1 & -x_{p+1} & -y_{p+1} & 0 & 0 & 0 \\ \\ 0 & 0 & 0 & -1 & -x_{p+q} & -y_{p+q} \end{bmatrix}_{q\times 6} \end{bmatrix}}_{A_{(p+q)\times 6}} \underbrace{\begin{bmatrix} \beta_0^i \\ \beta_1^i \\ \beta_2^i \\ \beta_0^j \\ \beta_1^j \\ \beta_2^j \end{bmatrix}}_{\cdot\beta_{6\times 1}} = \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \\ \varepsilon_{p+q} \end{bmatrix}}_{\varepsilon_{(p+q)\times 1}} \tag{3}$$

The model parameters $\boldsymbol{\beta}$ for each SAR frame can be estimated by minimizing the sum of squared residuals $\varepsilon = \left[\varepsilon_1, \ldots, \varepsilon_p, \ldots, \varepsilon_q\right]^T$, using the least-squares method. The modeled bias is finally subtracted from the observed velocities on a frame-by-frame basis. For nonlinear long-wavelength bias, the higher-degree polynomial can be employed.

4. Methodology

4.1. Modeling horizontal velocity

For wide-area scenarios, the presence of tectonic motion will increase the difference of LOS velocity in the overlap region between two adjacent SAR tracks. Here we propose incorporating a plate motion model and the CORS network in the determination of the vertical velocity component. Using the horizontal motion of tectonic plates across the Earth's surface (Goudarzi et al., 2014),

where $[v_n, v_e]_p^T$ denote the tectonic velocity at a position p along the northern and eastern directions in a local geodetic Cartesian coordinate system. r_e is the Earth's radius. ϕ and λ are the spherical latitude and longitude. Using Eq. (4), we can calculate the regional horizontal velocity at an arbitrary position, given the Euler vector of the plate $\left[\omega_x, \omega_y, \omega_z\right]^T$. To do this, the Euler vector at the NCP should be inverted. Some widely-used international plate motion models also provide Euler vectors for large-scale tectonic plates, e.g., NNR-NUVEL-1 A model for the Eurasian plate, but they cannot accurately characterize the plate motion at regional scales (Ren et al., 2020; Wang and Shen, 2020). Therefore, we initially used the CORSs with the crustal secular velocity solution published by Wang and Shen (2020) to invert the regional Euler vector $\left[\omega_x, \omega_y, \omega_z\right]^T$ by a least-squares, and then calculated the tectonic velocity using Eq.(4) at each point of interest. For each InSAR frame, we transformed the absolute tectonic velocity in the ITRF2008 reference frame into a relative reference by subtracting the tectonic velocity value at the location of the InSAR local reference point. Then the InSAR

vertical velocity at an arbitrary position is determined by combining Eq. (1) and Eq. (4),

$$v_{UD} = \frac{(v_{LOS} - v_{NS}\sin\theta\sin\alpha + v_{EW}\sin\theta\cos\alpha)}{\cos\theta}, v_{EW} \approx v_e \text{ and } v_{NS} \approx v_n \quad (5)$$

Eq. (5) constrains the horizontal deformation field of InSAR through tectonic motion model, thereby reducing the bias in projected vertical velocities that would occur under the assumption of zero horizontal motion.

4.2. Modeling systemic errors in velocity field and estimation

The velocity fields in different SAR frames are affected by varying error patterns, making it infeasible to use a single model to correct bias uniformly across frames. A more suitable strategy is to assign different models to different frames based on their specific bias characteristics. In this context, the fixed linear model in Eq. (3) can be extended to the more flexible form in Eq. (6), incorporating multiple model candidates. Assuming the Δ_p^i represents the bias to be corrected at position p in the ith frame, the model can be defined as,

$$\Delta_p^i = \begin{cases} \beta_0^i & (a) \\ \beta_0^i + \beta_1^i x_p + \beta_2^i y_p & (b) \\ \beta_0^i + \beta_1^i x_p + \beta_2^i y_p + \beta_3^i x_p y_p & (c) \\ \beta_0^i + \beta_1^i x_p + \beta_2^i y_p + \beta_3^i x_p y_p + \beta_4^i x_p^2 + \beta_5^i y_p^2 & (d) \end{cases} \quad (6)$$

where β_n^i $(n = 0, 1, 2, \ldots, 5)$ represents the parameters of four sub-models for the ith frame, and x_p and y_p again are the coordinates of the pth tie point. The objective is to select the most appropriate sub-model for each SAR frame from Eqs. (6a) to (6d), based on a suitable model selection criterion.

Following the structure of Eq. (3), if the constant model in Eq. (6a) is the optimal sub-model for the ith frame and the linear model in Eq. (6b) is optimal for the jth frame, then the original linear model for the ith frame is replaced by a constant term, and the linear system is modified accordingly,

$$\underbrace{\begin{bmatrix} \begin{bmatrix} v_1^i - v_1^j \\ v_2^i - v_2^j \\ \\ v_p^i - v_p^j \end{bmatrix}_{p\times 1} \\ \begin{bmatrix} v_{p+1}^i - v_{p+1}^{i,GNSS} \\ \\ v_{p+q}^j - v_{p+q}^{j,GNSS} \end{bmatrix}_{q\times 1} \end{bmatrix}}_{v_{(p+q)\times 1}} + \underbrace{\begin{bmatrix} \begin{bmatrix} -1 & 1 & x_1 & y_1 \\ -1 & 1 & x_2 & y_2 \\ & & & \\ -1 & 1 & x_p & y_p \end{bmatrix}_{p\times 4} \\ \begin{bmatrix} -1 & 0 & 0 & 0 \\ & & & \\ 0 & -1 & -x_q & -y_q \end{bmatrix}_{q\times 4} \end{bmatrix}}_{A_{(p+q)\times 4}\cdot\beta_{4\times 1}} \begin{bmatrix} \beta_0^i \\ \beta_0^j \\ \beta_1^j \\ \beta_2^j \end{bmatrix} = \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \\ \varepsilon_{p+q} \end{bmatrix}}_{\varepsilon_{(p+q)\times 1}} \quad (7)$$

When the weight matrix $\mathbf{w}$ is known, the solution of parameter vector $\boldsymbol{\beta}$ can be found by minimizing the weighted sum of squared residuals using the least-squares adjustment,

$$\hat{\beta} = \left(A^T \mathrm{w} \mathrm{A}\right)^{-1} A^T \mathrm{w} \mathrm{v} \quad (8)$$

where $\mathbf{w}$ is a $(p+q)\times(p+q)$ matrix with known positive numbers on the diagonal and 0 elsewhere,

$$w = diag\left(\left[\frac{1}{\sigma_{i,1}^2 + \sigma_{j,1}^2}, , \frac{p/q}{\sigma_{j,q}^2 + \sigma_{j,GNSS,q}^2}\right]_{p+q}\right) \quad (9)$$

where $\sigma_{i,p}^2$, $\sigma_{j,p}^2$ and $\sigma_{j,GNSS,q}^2$ correspond to the uncertainty in InSAR and GNSS velocities, respectively. The uncertainty of InSAR velocity is the sum of the InSAR-derived uncertainty and the tectonic velocity uncertainty, which can be derived from InSAR time-series inverse (Ou et al., 2022) and the plate motion model, respectively. The uncertainty of GNSS data is provided by CMONOC. p/q is used to balance the ratio between the number of InSAR and GNSS observations (Ou et al., 2022).

Since the optimal sub-model for each SAR frame is unknown a prior, we can use K-fold cross-validation in machine learning that partitions the dataset into multiple subsets to iteratively train and test the candidates until the best one is found. Specifically, given N frames and M candidate sub-models per frame, there are M^N possible sub-model combinations. For each combination, we partition the velocities of p InSAR tie points and q GNSS control points into K subsets (folds). Each subset $\mathbf{v}_k^T, k = 1, 2, \ldots, K$ has a sample size of about $(p+q)/K$. The validation procedure is as follows:

1) For a given sub-model combination, train the model using K-1 subsets $\left[v_1^T, , v_{k-1}^T, v_{k+1}^T, v_K^T\right]$ and compute the root-mean-square error (*rmse*) on the held-out subset $\mathbf{v}_k^T$,

$$rmse_k = \frac{\|A_k \hat{\beta}_k - v_k\|}{\sqrt{(p+q)/K}} \quad (10)$$

where $\mathbf{A}_k$ is a design matrix similar to the one in Eq. (7). Both $\mathbf{A}_k$ and

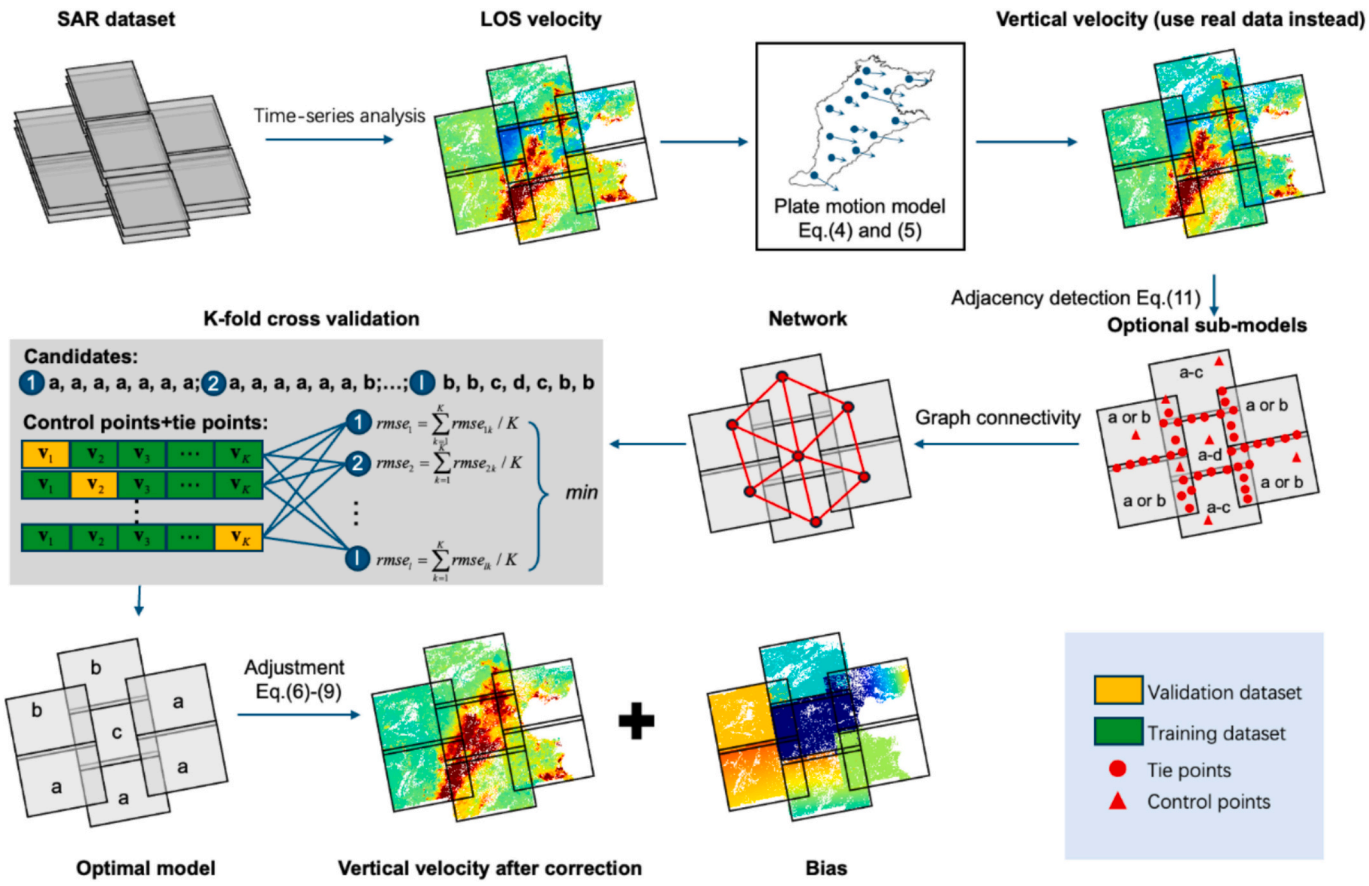


Fig. 3. Pictorial example of machine learning-based adjustment for wide-area interferometry.

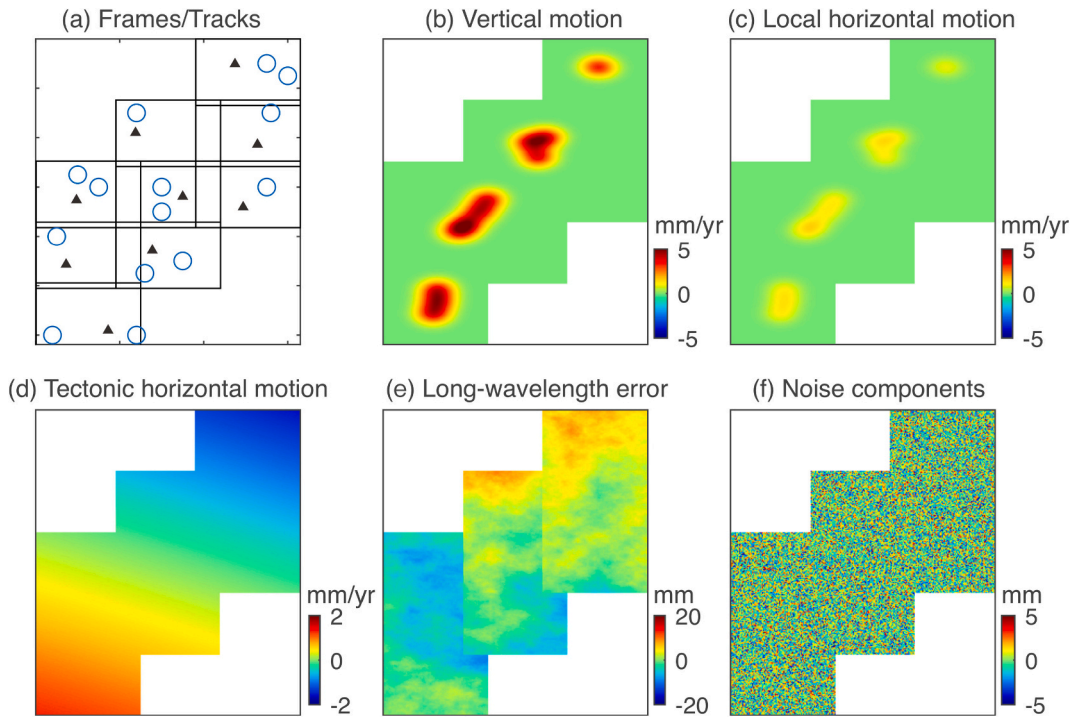


Fig. 4. (a) SAR coverage. The triangle in black shows the local reference points and the blue circles are control points. (b) Vertical motion. (c) Local horizonal motion. (d) Regional horizontal motion. (e) Long-wavelength bias and (f) stochastic Gaussian noise. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

estimate $\widehat{\boldsymbol{\beta}}_k$ in Eq. (10) depend on the $\left[v_1^T,,v_{k-1}^T,v_{k+1}^T,v_K^T\right]$ and candidates to be selected for all frames.

2) Repeat step 1) K times, using each subset once as the test set while the remaining K-1 subsets serve as training data.
3) Compute the averaged *rmse* across all K folds as the performance metric for this combination.
4) Repeat steps 1) through 3) for all M^N combinations, then select the combination with minimum averaged *rmse* as optimal.

4.3. Algorithm development for machine learning-based adjustment

We first design a flexible strategy to reduce the number of combinations to address the exponential growth of computation time. In Fig. 2, the adjacency of SAR frames can be expressed as an undirected graph where each frame represents a node and the overlap region represents an edge. The higher degree of graph node (i.e., the number of edges connected to each node) generally increases the reliability of block adjustment because of redundant observations. However, this is not true for high-order polynomials as shown in Fig. 2b, where the top left corner of the frame in blue is out of control. Therefore, the sub-models in Eq. (6) should be selected according to the distribution of tie points and overlap regions. To do this, we divide the area around each node in eight partitions B_i in rad (Fig. 2d), each of which is $\pi/4$. By calculating relative positions of tie points $\left(x_p,y_p\right)$ to the central of ith frame $\left(x_c^i,y_c^i\right)$, the orientation of tie points can be defined as,

$$\angle\theta_p = atan2\left(\frac{y_p - y_c^i}{x_p - x_c^i}\right) + \pi \tag{11}$$

By collecting the partitions of which $\angle\theta_p$ is a member, a set $B = \underset{i}{\cup} B_i$, $i \le 8$, quantifies the distribution of tie points with respect to the central frame by a positive angle. For example, the blue frame in Fig. 2a has only one overlap region at the bottom, and the tie points are in two consecutive partitions, we have $B = \pi/2$. Likewise, we have $B = \pi$ for the blue frame in Fig. 2b and $B = 3\pi/2$ for the blue frame in Fig. 2c. The following criterion is defined immediately for the model selection: if $B \le \pi/2$, only constant model in Eq. (6a) is considered; if $\pi/2 < B \le \pi$, Eq. (6a-b) is suitable; if $\pi < B \le 3\pi/2$, Eq. (6a-c) is included; if $B > 3\pi/2$, arbitrary model can be deployed. When dense GNSS observations are available, the extension to both tie points and control points is straightforward.

Fig. 3 illustrates the workflow of the machine learning-based InSAR adjustment. Following time-series analysis, the LOS velocity field of each frame is first converted to the vertical component by decomposing the horizontal displacements using the plate motion model described in Eqs. (4) and (5). The potential sub-models for each frame are then determined according to Eq. (11). Once the graph connectivity is established, the design matrix in Eq. (7) is constructed for each candidate with its specified sub-models. Using K-fold cross-validation, the average performance of each candidate is assessed across the K validation datasets ($K = 10$ is used throughout this paper). Finally, the optimal candidate corresponding to the minimum *rmse* value is selected to model the bias in the vertical velocity field using Eq. (8) and (9). The corrected velocity field is obtained by mosaicking the individual frame velocities over the wide-area domain after removing their respective biases.

5. Results

5.1. Synthetic data

The performance of the presented approach was evaluated by synthetic data from three tracks, each comprising three frames. The along-track and cross-track areas refer to Sentinel-1 geometry, which has a spatial resolution of 1 km after down-sampling within the 250-km swath and 12-day revisit cycle. We simulated 98 time-series unwrapped phases for each frame according to vertical velocity in Fig. 4b with the interval [−5 5] mm/yr, local horizontal velocity in Fig. 4c with a maximum of 5 mm/yr, regional horizontal velocity in Fig. 4d by the Euler vector based on the GNSS data over the NCP, long-wavelength bias in the interval [−20,20] mm in Fig. 4e, and stochastic noise in the interval [−5, 5] mm in Fig. 4f. The same atmospheric screen between consecutive frames is assumed due to similar imaging times. We calculated LOS velocity and then referenced them to the local reference point in black frame by frame in Fig. 4a. The control points in blue cycle were used for block adjustment.

Fig. 5 shows estimated vertical velocities and their dispersions relative to the reference truth. Spatial inconsistencies and large biases are clearly visible in the original velocity maps (a1, b1). After

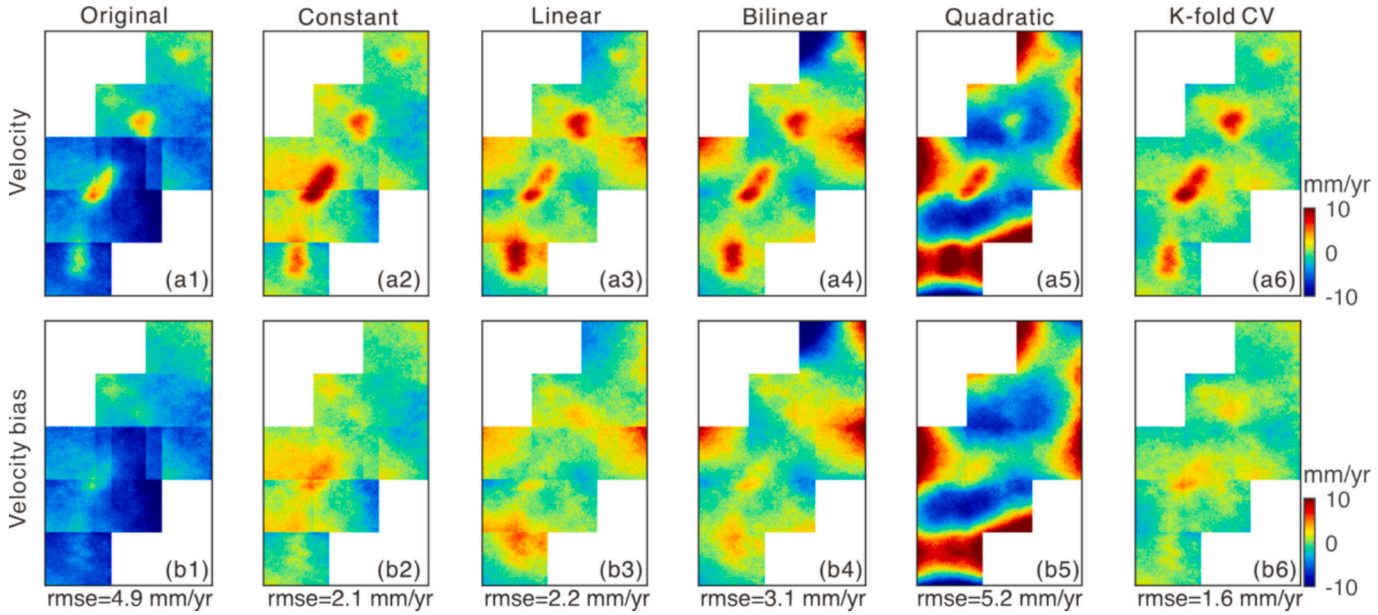


Fig. 5. The estimated velocity (the first row) and corresponding bias between the truth and the estimated velocities (the second row) under adjustments with different models.

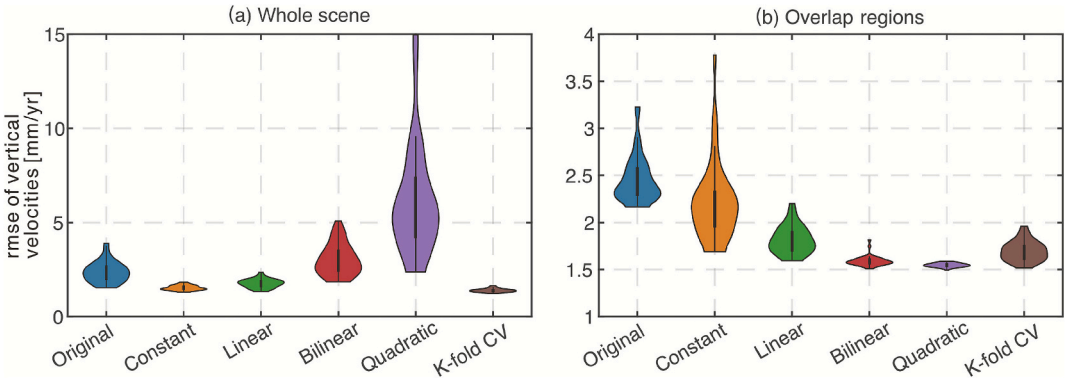


Fig. 6. Accuracy assessment for (a) the whole and (b) the overlap regions, using rmse between the estimated wide-area velocity field and the ground truth.

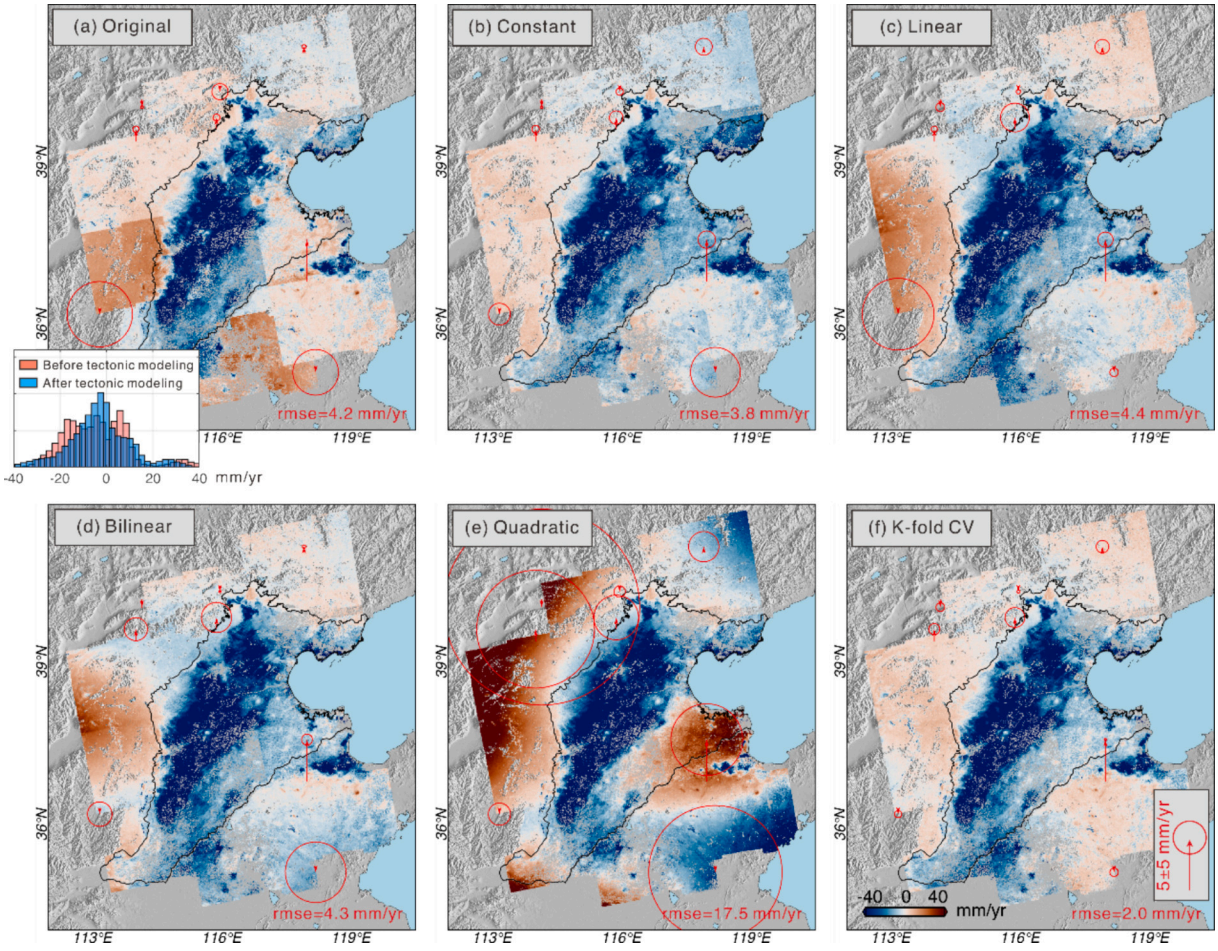


Fig. 7. Vertical InSAR velocities over the NCP from different methods: (a) original velocity with tectonic model constraints, (b)-(e) velocity after block adjustment using constant, linear, bilinear model, and quadratic models, and (f) *K*-fold cross-validation. Subplot in (a) shows histograms of velocity differences in overlap regions without (red) and with (blue) tectonic model constraints. Red arrows indicate the GNSS vertical velocity, and red circles represent the absolute differences between the GNSS and InSAR velocities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

adjustment, the constant and linear models (a2-a3, b2-b3) reduce biases but introduce data inconsistencies. In contrast, the bilinear and quadratic models (a4-a5, b4-b5) improve consistency but yield larger biases near the image borders. This behavior is expected, as higher-order polynomials better fit the training data in the overlap regions but tend to overfit areas distant from control and tie points. In comparison, the machine learning-based adjustment (a6, b6) effectively balances bias and consistency by adaptively determining polynomial orders according to the spatial distribution of control and tie points.

Quantitatively, we evaluated the overall performance of different methods using 50 Monte Carlo simulations, where both long-wavelength bias and stochastic errors were varied. Two scenarios were analyzed: one for the entire scene (Fig. 6a) and the other for the overlap regions (Fig. 6b).

The adjustments based on bilinear and quadratic models show the lowest accuracy for the entire scene in Fig. 6a, but the highest accuracy in the overlap regions in Fig. 6b. This is similar with the results in Fig. 5b4 and b5, where polynomials with higher order generally perform

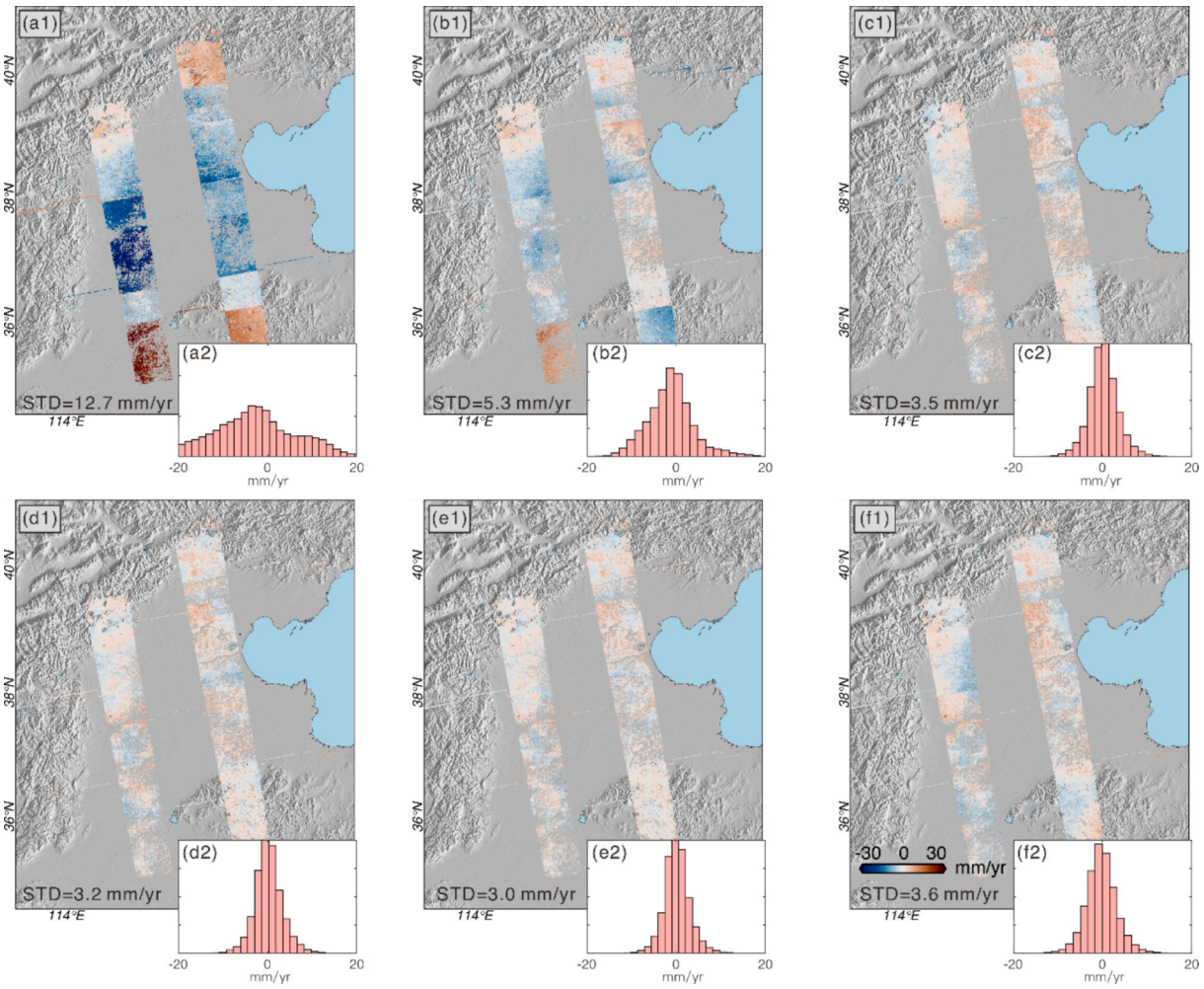


Fig. 8. Vertical velocity bias in overlap regions and corresponding statistical distributions of bias for different adjustment methods.

well on training data in overlap regions but tend to overfit at the regional edges. In contrast, the constant and linear models display higher accuracy for the entire scene in Fig. 6a but relatively lower accuracy in the overlap regions in Fig. 6b, as these simpler models are unable to fit more complex error patterns. The machine learning-based approach provides the best overall accuracy, with a *rmse* value of approximately 1.6 mm/yr. This highlights the limitations of empirical model selection, as *K*-fold cross-validation effectively balances model complexity and avoids overfitting.

5.2. Real data

To estimate the Euler vector more accurately over the NCP, our regional horizontal velocity solution integrates both the CORS data and the crustal secular velocity data published by Wang and Shen (2020). The histogram in Fig. 7a illustrates the vertical velocity dispersions between consecutive tracks in overlap regions, with and without tectonic modeling, where all InSAR-derived velocities were corrected to an absolute reference using individual GNSS observations within each frame. For the original velocity projected from the LOS into vertical direction, the standard deviation of dispersion reaches 14.3 mm/yr, which decreases to 12.7 mm/yr after removing the horizontal tectonic components. The spatial patterns of individual velocity fields are visually compared in Fig. S2 of the supplementary material.

During the block adjustment, we used 12 CORSs, represented by red triangles in Fig. 1b, to constrain the model parameters during the inversion. The remaining 8 CORSs, shown as black triangles, were reserved for validation. All corrected velocity fields, including the original one without block adjustment, are presented in Fig. 7. Similar to the simulation in Fig. 5, the constant and linear models lead to spatial discontinuities. In contrast, unpredictable biases estimated by the bilinear and quadratic models are widely distributed along the frame borders.

The bias behavior can be further quantified by evaluating the velocity dispersions within overlap regions and between InSAR and GNSS measurements in Fig. 8. Compared with the larger bias in Fig. 8a1 and 8a2, the subtle differences observed in histograms (c1-f1) imply that higher-order polynomial models are sufficient to eliminate biases within the overlap regions. However, in non-overlap areas, the dispersions of GNSS sites distributed along the frame borders (highlighted by red circles in Fig. 7) confirm that higher-order polynomials tend to overfit. Consistent with the conclusions drawn from the simulation in Fig. 5, the machine learning-based adjustment achieves a better balance between velocity biases in overlap and non-overlap regions. The *rmse* statistics between eight GNSS and InSAR velocities further support this conclusion: the *K*-fold cross-validation yields the smallest *rmse* of 2.0 mm/yr, whereas the *rmse* values for the constant, linear, bilinear, and quadratic models are 3.8, 4.4, 4.3, and 17.5 mm/yr, respectively.

6. Land subsidence in the NCP between 2014 and 2022

Fig. 9 shows the spatial distribution of subsidence and statistics with respect to land cover and hydrological materials. Overall, more than 63 % of the area, approximately 88, 200 km^2, experienced land subsidence

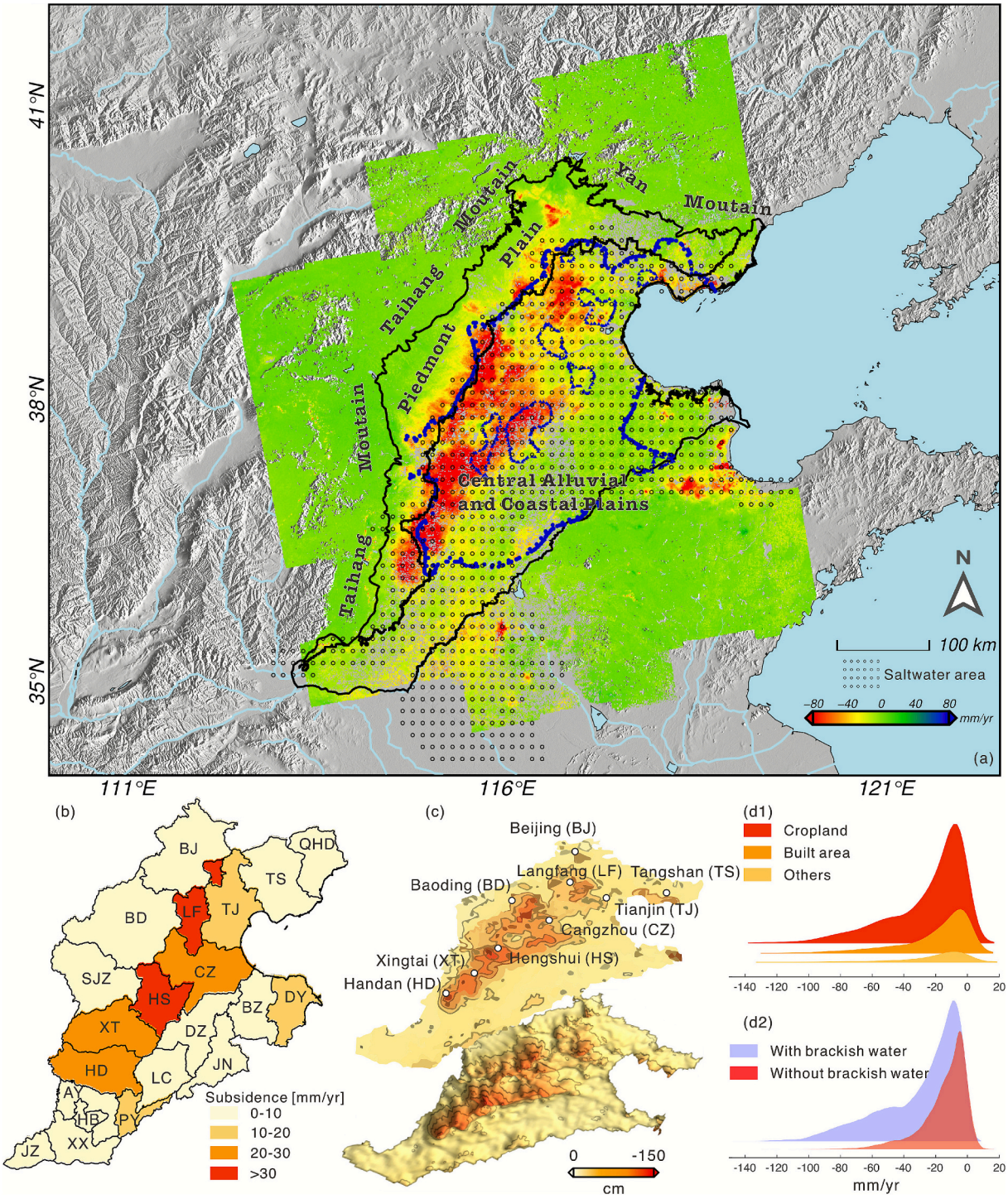


Fig. 9. Land subsidence across the NCP based on Sentinel-1 InSAR measurements. (a) wide-area land subsidence velocity field; The black line marks the boundaries of the NCP, while the blue dashed line outlines the boundaries of the confined groundwater depression cones. The textured circles indicate areas overlain by brackish water aquifers. (b) Annual subsidence share by city (abbreviations in Table S1). (c) Cumulative subsidence from 2014 to 2022 and corresponding 3D view. (d) Velocity statistics by (d1) land covers and (d2) brackish groundwater presence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

greater than 10 mm/yr between October 2014 and December 2022. An area of over 56, 882 km^2 has subsided more than 20 mm/yr, with the majority located in the compound groundwater cone of depression, highlighting the impact of deep groundwater exploitation on surface deformation. Areas with subsidence exceeding 30 mm/yr account for 28 % of the total area of the NCP. In comparison, surface subsidence in the piedmont plain is relatively minor due to the semi-confined aquifers, low exploitation intensity in deep groundwater (primarily from unconfined aquifers), the favorable lateral recharge conditions and stronger sediment systems (Guo et al., 2015), and the presence of stronger sediment systems. The large-scale subsidence observed in the textured area of Fig. 9a indicates overexploitation from confined and semi-confined aquifers beneath a brackish water aquifer in the central alluvial and coastal plains.

In Fig. 9b, our results revealed that land subsidence is occurring in many cities, with notable variability across different regions. This is particularly evident in the central alluvial plain, where five cities experienced annual subsidence exceeding 20 mm/yr. Ranked in descending order of subsidence magnitude, these cities are Hengshui (HS, 58.2 mm/yr), Langfang (LF, 35.5 mm/yr), Xingtai (XT, 29.6 mm/yr), Cangzhou (CZ, 28.5 mm/yr), and Handan (HD, 26.7 mm/yr). Additionally, three other cities, including Tianjin (TJ, 20.0 mm/yr), Puyang (PY, 19.1 mm/yr), and Dongying (DY, 17.6 mm/yr), experienced subsidence velocity greater than 10 mm/yr. In some hotspot areas reported in previous studies (Dong et al., 2023; Wang, Feng, et al., 2024), such as Beijing (BJ) and Dezhou (DZ), subsidence velocities have been reduced to 6.2 mm/yr and 8.8 mm/yr, respectively.

Fig. 9c shows the cumulative subsidence. Severe subsidence has resulted in the formation of numerous subsidence funnels, which have merged to form larger compound funnels, covering an area of approximately 24,000 km^2 across the central alluvial plain (Guo et al., 2015). We found that the maximum subsidence within this compound funnel occurs in Hengshui and Xingtai, reaching 1.2 m and 1.1 m, respectively. However, the greatest magnitude of cumulative subsidence is detected in Dongying (DY), far from the central plain, where it reaches 2 m, driven by local underground mining activities.

We further illustrate the indicative role of land cover and brackish water aquifer distribution on groundwater-induced subsidence, as shown in Fig. 9d. The average annual subsidence over cropland, built-up areas, and other land classes is 24.4 mm/yr, 19.3 mm/yr, and 17.8 mm/yr, respectively, based on the land cover raster product with a 100 m resolution from the Copernicus Land Monitoring Service (https://land.copernicus.eu/en). Urban water supply and irrigation serve as the primary drivers of groundwater extraction in the North China Plain (Kinzelbach et al., 2022), with extensive pumping resulting in significant subsidence across both urban and cropland areas. Additionally, the influence of brackish water distribution is even more pronounced: the average velocity in areas overlying brackish water aquifers is more than double that in areas without brackish water, at 29.0 mm/yr compared to 12.3 mm/yr. Compared to areas without brackish water, where communities can utilize both shallow and deep groundwater supplies, regions overlying brackish water aquifers rely more heavily on deep groundwater from confined aquifers due to superior water quality, consequently experiencing more severe subsidence.

7. Discussion

7.1. Computational performance

One of the novelties of this paper is the joint use of different models to minimize long-wavelength bias by machine learning. Model selection can be time-consuming and negatively impacts computational efficiency. We present an efficient strategy that narrows the pool of selectable candidates by exploring the distribution of overlap regions relative to the central frame. For four sub-models in Eq. (6) and 11 Sentinel-1 SAR frames, we reduced over 4 million candidates (4^{11}) to 62,208, resulting in a 67.4-fold increase in efficiency.

The execution time of K-fold cross-validation for each candidate depends on the value of K and the design matrix $\mathbf{A}$ in Eq. (8). In this study, we selected $K = 10$ and reduced the size of $\mathbf{A}$ by down-sampling the image to a 1 km grid. Consequently, for the SAR coverage of about 300,000 km^2, the K-fold cross-validation takes 0.0045 s per candidate on a MATLAB platform with an Intel® Core™ i9-13900K @ 5.80 GHz processor. The computation time for cross-validation is approximately 62208*0.0045 s = 279.93 s. Despite this additional computational overhead, the K-fold cross-validation improves accuracy (Fig. 7) while reducing memory consumption compared with high-order polynomials (see Figs. S4 and S5 in the supplement material). This efficiency gain arises because higher-order polynomials at full-resolution adjustment require design matrices with larger size. Adaptive model selection avoids unnecessary memory allocation by preventing uniform application of complex models where simpler ones suffice.

7.2. Error propagation and uncertainty analysis

The uncertainty of corrected velocity field after adjustment can be evaluated by the covariance matrix of the model parameters $\widehat{\boldsymbol{\beta}}$ in Eq. (8). According to the error propagation law, the covariance matrix of $\widehat{\boldsymbol{\beta}}$ is $\mathbf{C}_{\widehat{\boldsymbol{\beta}}} = \sigma^2\left(\mathbf{A}^T\mathbf{w}\mathbf{A}\right)^{-1}$.The posteriori variance of unit weight σ^2 is computed from,

$$\sigma^2 \approx \frac{(\mathbf{A}\widehat{\boldsymbol{\beta}} - \mathbf{v})^T\mathbf{w}(\mathbf{A}\widehat{\boldsymbol{\beta}} - \mathbf{v})}{p + q - n_{\widehat{\boldsymbol{\beta}}}} \tag{12}$$

where $p + q$ denotes the number of tie points and control points, and $n_{\widehat{\boldsymbol{\beta}}}$ is the number of model parameters in the whole scene. Note that the parameter number $n_{\widehat{\boldsymbol{\beta}}_i}^{(i)}$ in each frame is different in Eq. (7), we have $n_{\widehat{\boldsymbol{\beta}}} = \sum n_{\widehat{\boldsymbol{\beta}}_i}^{(i)}$. The total variance of the corrected InSAR velocity field can be expressed as the sum of the variance of original InSAR velocity and the variance of the model prediction.

In practical solution, we first indexed all frames in ascending order based on acquisition time. Suppose that the quadratic model is the optimal model in the ith frame, we have $n_{\widehat{\boldsymbol{\beta}}_i}^{(i)} = 6$. Subsequently, the submatrix $\mathbf{C}_{\widehat{\boldsymbol{\beta}}}^i$ was extracted from $\mathbf{C}_{\widehat{\boldsymbol{\beta}}}$ by selecting rows and columns ranging from $m + 1$ to $m + 6$, where m is the number of parameters in i-1 frames, $m = \sum_{l=1}^{i-1} n_{\widehat{\boldsymbol{\beta}}_l}^{(l)}$. Given a position p with coordinates (x_p, y_p) in the ith frame, the uncertainty of corrected velocity $\widehat{\sigma}_{i,p}$ is,

$$\begin{cases} C_{\widehat{\beta}}^i = C_{\widehat{\beta}}(m+1:m+6, m+1:m+6) \\ \left(\sigma_{i,p}^{\text{model}}\right)^2 = \left[1, x_p, y_p, x_p y_p, x_p^2, y_p^2\right] C_{\widehat{\beta}}^i \left[1, x_p, y_p, x_p y_p, x_p^2, y_p^2\right]^T \\ \widehat{\sigma}_{i,p} = \sqrt{\left(\sigma_{i,p}^{\text{model}}\right)^2 + \sigma_{i,p}^2} \end{cases} \tag{13}$$

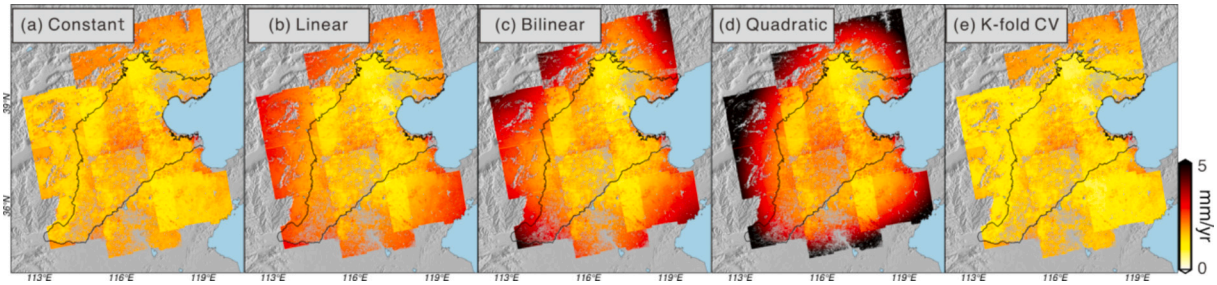


Fig. 10. Uncertainties of InSAR velocity field over the NCP estimated by different methods: velocity after block adjustment using (a) the constant model, (b) the linear model, (c) the bilinear model, (d) the quadratic model and (e) the K-fold cross-validation.

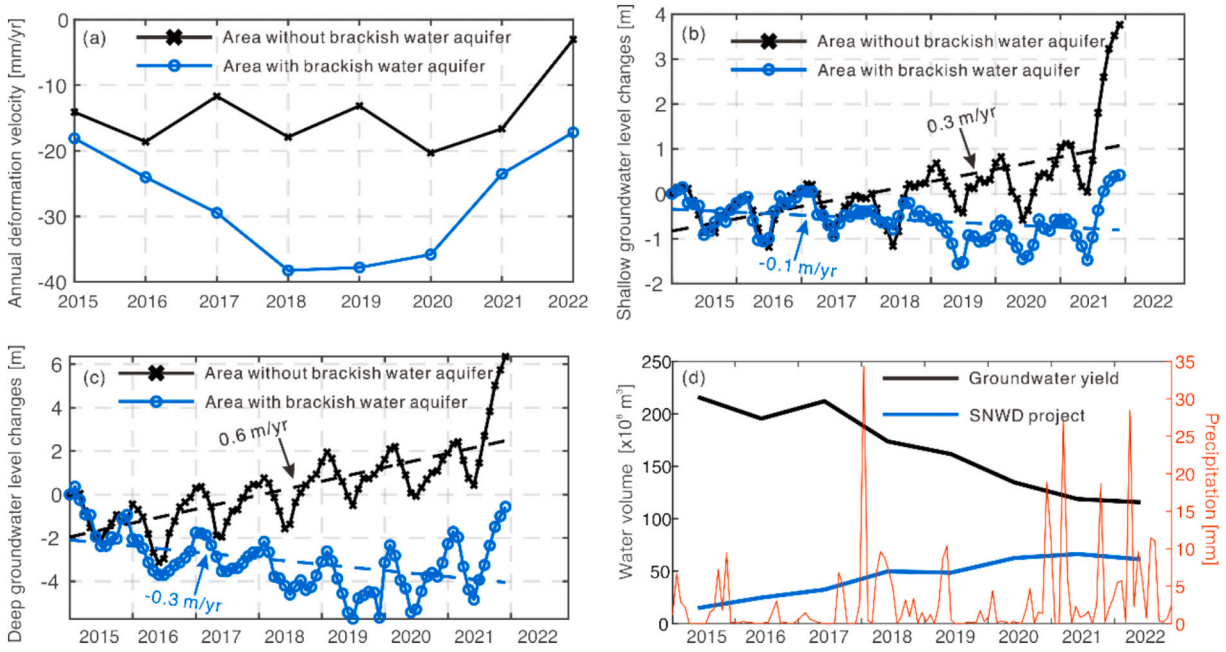


Fig. 11. Time-series variables in the NCP from the end of 2014 to 2022, showing (a) annual subsidence, (b) shallow groundwater levels, (c) deep groundwater levels, and (d) statistics on groundwater withdrawals, water transfers from the SNWD project, and precipitation.

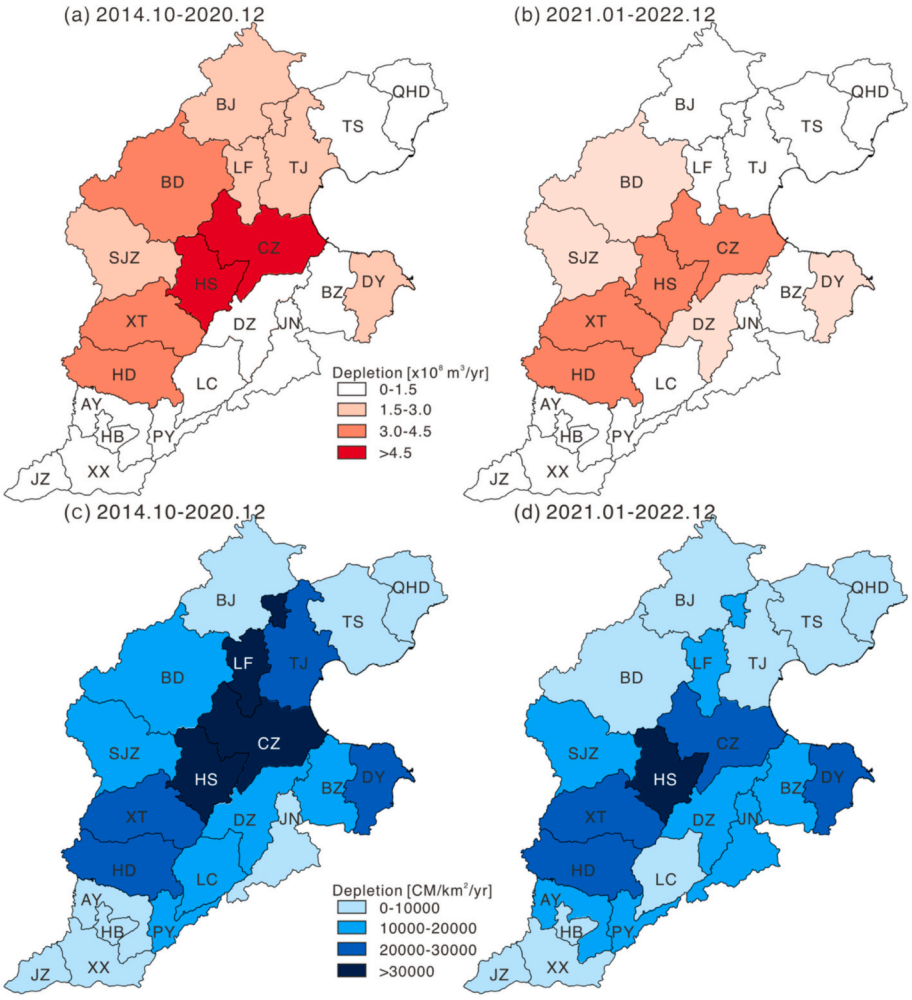


Fig. 12. Groundwater storage loss related to land subsidence in confined aquifer systems in the NCP. The contribution of each city to annual groundwater depletion for the periods of 2014.10–2020.12 (a) and 2021.01–2022.12 (b) is presented. (c-d) Annual groundwater depletion per square kilometer in different cities.

where $\left(\sigma_{i,p}^{\text{model}}\right)^2$ is the variance of model prediction and $\sigma_{i,p}^2$ is the variance of original InSAR velocity that can be obtained from InSAR time-series analysis.

The uncertainty analysis in the NCP is illustrated in Fig. 10. After applying block adjustment based on linear, bilinear, and quadratic models, we observed that errors extend toward the frame borders, resulting in increased uncertainties of 1.8 mm/yr, 1.9 mm/yr, and 2.3 mm/yr, as shown in Fig. 10b-d, respectively. Factors such as the use of inappropriate models to fit the data, the ill-conditioned matrix caused by collinear observations, and overfitting contribute to varying degrees of error propagation. The machine learning-based approach seeks to identify the optimal model that minimizes errors globally, reducing the uncertainty to 1.4 mm/yr in Fig. 10e.

7.3. Analysis of annual temporal evolution

Fig. 11a presents the time series of annual average subsidence. Compared to interannual fluctuations in subsidence velocity in areas without brackish aquifers, regions with brackish aquifer coverage experienced more pronounced subsidence, reaching a peak of 38.5 mm/yr in 2018. In Fig. 11b and c, annual average groundwater level changes in both shallow and deep layers (black line) show a long-term upward trend over the piedmont plain, indicating better interaction between the aquifer system and natural or artificial recharge. In the central alluvial and coastal plains, a clear trend of groundwater level drawdown (blue line) is observed, indicating the challenges of freshwater replenishment, particularly regarding how water infiltrates into the deep aquifer and the turnover time.

It is worth noting that starting from 2021, subsidence across the entire NCP has alleviated. Specifically, average subsidence in the piedmont plain tended to stabilize at 3 mm/yr by 2022, while in the central alluvial and coastal plains, which have brackish water coverage, the subsidence velocity returned to 2015 levels, reaching 18.8 mm/yr. This phenomenon is illustrated in Fig. 11d, where we quantified groundwater yield, the water transfer volume of the SNWD project, and precipitation in the NCP from the end of 2014 to December 2022. The continuous decrease in groundwater withdrawals (approximately 11.7 billion m^3/yr after December 2020), the increase in water transfer volume (about 6.4 billion m^3/yr after December 2020), and the rise in annual precipitation (569 mm/yr and 832 mm/yr before and after December 2020) are the primary drivers of the water level rise post-2020. Increased water flowing into the pore spaces of aquifer sediments reduces the effective stresses on the grains of the porous medium, leading to aquifer expansion.

In the piedmont plain, sediments primarily composed of sand and gravel exhibit higher hydraulic conductivity and better permeability, allowing surface rebound to respond more promptly to groundwater recovery. However, for unconsolidated aquifer-aquitard systems with thick clay-rich deposits, the compressibility of the aquitard during expansion is limited, and the time lag between water level changes and surface deformation can vary significantly from years to decades or more (Bundschuh and Arriaga, 2010; Su et al., 2023). Therefore, we speculate that the primary reason for the slowdown in subsidence in the central alluvial and coastal plains is the long-term restrictions on groundwater withdrawals (black line in Fig. 11d), while the abnormal increase in rainfall has indirectly reduced the demand for groundwater pumping for irrigation.

7.4. Confined groundwater storage loss associated with land subsidence in the NCP

The data interpretation in Section 6 demonstrates that the compaction process in aquitards and aquifers, caused by excessive groundwater pumping, is the primary contributor to wide-area land subsidence. Groundwater extraction from confined and semi-confined aquifers is the dominant cause of land subsidence. According to inverse theory of groundwater presented by Smith and Majumdar (2020), the confined groundwater storage loss ΔS can be equivalent to the volume of land subsidence,

$$\Delta S = \Delta h \bullet S_s \bullet A \bullet b \approx \Delta h \bullet \frac{\Delta b}{b \bullet \Delta h} \bullet A \bullet b = A\Delta b \tag{14}$$

where Δh denote the confined groundwater level changes. S_s is the specific storage coefficient. A is the area and b is the thickness of the aquifer system. Δb represents aquifer compaction, which corresponds to the land subsidence measured by InSAR in this study. The Eq. (14) demonstrates that the product of InSAR-measured subsidence and its corresponding area in each pixel serves as a direct proxy for water volume released through aquifer compression. We also calculated elastic unloading deformation caused by water mass changes (Text S1 in supplementary material), which shows maximum velocities less than 1.3 mm/yr (Fig. S3). Hence, its impact on our estimation of water loss is essentially negligible.

Overall, the NCP experienced a confined aquifer storage loss of 24.9 billion m^3 over an 8-year period, corresponding to an annual loss of approximately 3.11 billion m^3. Between October 2014 and December 2020, the loss amounted to 20.2 billion m^3, accounting for 81 % of the total loss, while the loss from January 2021 to December 2022 was 4.7 billion m^3, representing 19 % of the total.

Analyzing the spatial distribution of groundwater loss at the city scale provides insights into the local changes in water source depletion. In Fig. 12a, it is clear that Hengshui (HS) and Cangzhou (CZ) experienced the highest water depletion stress, with an annual groundwater depletion of 0.58 billion m^3/yr and 0.51 billion m^3/yr, respectively, accounting for 28 % of the total loss in the NCP from the end of 2014 to December 2020. Baoding (BD), Xingtai (XT), Handan (HD) ranked second-level, totally contributing 25 %.

Normalized groundwater loss for the area of each city reveals the water resource depletion per unit of land across different regions (Fig. 12c). Hengshui (HS), Langfang (LF), and Cangzhou (CZ) rank first with annual groundwater losses of 55,000, 33,000, and 30,000 cubic meters per square kilometer, respectively. It is worth noting that although Xingtai (XT) and Handan (HD) are not the most severely affected cities in terms of total or normalized groundwater loss, they deserve particular attention due to two groundwater depression cones as shown in Fig. 9, which are primarily distributed in localized areas in the eastern parts of these cities.

Compared to the water stress levels before December 2020, groundwater depletion in most cities significantly declined from

Table 2
Comparison of subsidence across different cities. Maximum, median, and mean values represent the maximum recorded, median, and mean vertical velocity within each city, respectively.

References	Study area	Time period	Subsidence	This Study
(Dong et al., 2023)	Beijing	2015–2021	~120 mm/yr (maximum)	126.4 mm/yr
(Ao et al., 2024)	Beijing	2015–2022	2.5 mm/yr (median)	1.0 mm/yr
(Xu et al., 2025)	Beijing	2015–2019	10.0 mm/yr (mean)	5.6 mm/yr
(Su et al., 2023)	Tianjin	2014–2022	~100 mm/yr (maximum)	104.3 mm/yr
(Ao et al., 2024)	Tianjin	2015–2022	14.7 mm/yr (median)	11.6 mm/yr
(Xu et al., 2025)	Tianjin	2015–2019	15.12 mm/yr (mean)	19.6 mm/yr
(Shi et al., 2022)	Langfang	2015–2020	122 mm/yr (maximum)	125.7 mm/yr
(Ao et al., 2024)	Shijiazhuang	2015–2022	1.7 mm/yr (median)	0.2 mm/yr

January 2021 to December 2022. Thirteen cities experienced groundwater loss of less than 0.15 billion m^3/yr. In particular, due to the combined effects of the SNWD project and local policies that replenish ecological water into rivers and lakes (Su et al., 2023), groundwater release in the aquifer systems of Beijing and Tianjin, two major metropolises in China, decreased to 0.000016 billion m^3/yr and 0.07 billion m^3/yr, respectively. This substantial reduction in groundwater loss reflects the positive outcomes of effective water management practices in addressing water shortages.

7.5. Comparison with existing studies

We compared our regional results with existing studies at the local scale over similar monitoring periods, as shown in Table 2. Three metrics (maximum, median, and mean velocities) from existing Sentinel-1-based studies covering four major cities in the NCP were collected for comparison. Overall, the differences between previous studies and our InSAR-derived rates remain at the millimeter level across all cities and metrics. The largest discrepancy is observed in Beijing, where the maximum velocity differs by 6.4 mm/yr in the Zhaoyang District. For Beijing, Tianjin, and Shijiazhuang, the differences in mean and median velocities are all below 5 mm/yr. These results demonstrate that our method can effectively reduce regional biases and remain consistent with localized observations. Moreover, the regional-scale perspective allows the identification of new insights into the spatial patterns of subsidence. For example, more severe subsidence tends to occur in areas dominated by cropland and regions with brackish groundwater resources (Fig. 9), which were not evident in localized studies.

In addition, this study extends previous localized investigations of confined groundwater storage changes in the NCP to a regional scale. For the first time, we estimate a regional confined groundwater storage loss of 24.9 billion m^3 during 2014–2022, enabling city-level comparisons of groundwater depletion. Our analysis identifies Hengshui, Cangzhou, and Langfang as the areas experiencing the most severe confined groundwater loss, underscoring the need for targeted water management and subsidence mitigation strategies in these cities.

8. Conclusion

Since 2014, a series of water resource management measures, including the SNWD project, water resource fees and taxes, and groundwater extraction bans, have been implemented to mitigate water scarcity and promote sustainable development in the NCP. In this paper, we quantified the effects of these measures on wide-area land subsidence at fine resolution, using SAR instruments. We presented a machine learning-based approach to map consistent land subsidence across large-scale scenarios. The technical novelty lies in the use of a specific model for each SAR frame to minimize long-wavelength biases in the velocity field, which vary from frame to frame, rather than relying on a unified model for all SAR frames, as is common in state-of-the-art approaches. Additionally, we incorporated GNSS data into the block adjustment process to mitigate the impact of horizontal motion from tectonic plates on variations in incidence angles in the overlap regions between adjacent SAR tracks. Using both synthetic and real Sentinel-1 data, we validated the effectiveness of the presented method against independent GNSS measurements. When compared to prevalent adjustments based on constant, linear, bilinear, and quadratic models, the accuracy is improved from 3.8 to 17.5 mm/yr to 2.0 mm/yr.

We then applied this approach to investigate land subsidence across the entire NCP from the end of 2014 to 2022. Several new insights were discovered: (1) about 63 % of the area experienced land subsidence, with over 56,882 km^2 subsiding by more than 20 mm/yr. The majority of subsidence occurred in the central alluvial and coastal plains, with the maximum cumulative subsidence reaching approximately 2 m, observed in Dongying; (2) the central alluvial and coastal plains was subject to the most severe annual subsidence in 2018, reaching 38.5 mm/yr. However, starting in 2021, land subsidence in the NCP has been significantly alleviated. By 2022, the average subsiding was approximately 3 mm/yr in the piedmont plain and 18.8 mm/yr in the central alluvial and coastal plains, respectively; (3) the NCP experienced a confined aquifer storage loss of at least 24.9 billion m^3. From October 2014 to the end of 2020, the loss amounted to 20.2 billion m^3, accounting for 81 % of the total loss. Between January 2021 and December 2022, the loss decreased to 4.7 billion m^3, representing 19 % of the total. The results also indicated that in some hotspot areas, such as Beijing and Tianjin, groundwater depletion is approaching cessation. From the perspective of wide-area InSAR technique, we may conclude that since 2021, China's water resource management policies have had a positive impact on the control of land subsidence in the NCP.

CRediT authorship contribution statement

Mi Jiang: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Zhou Wu:** Visualization, Validation, Formal analysis, Data curation. **Xudong Wang:** Visualization. **Lin Bai:** Writing – review & editing. **Zhiwei Li:** Writing – review & editing, Supervision, Funding acquisition. **Zhong Lu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2025.115226.

Data availability

Data will be made available on request.

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